

Introduction

In this day and age, advanced data software is dominant in so many fields of science and other industries. However, creating good graphics that deliver the correct information remains a distinct challenge. Modern tools are accessible by anyone and often misused - especially in the context of using R. The result of these poorly designed graphics is 'chartjunk' - or over-designed graphics that force viewers to sift through useless information to find basic metrics and data. Steve Fiegdor establishes a framework for creating graphics in 'The Elements of Graphing Data Well', synthesizing the visual perception theories of industry legends to provide a sort of roadmap to great data design.

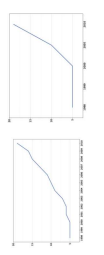
The Basics of Aesthetics in a Graph

The first three commandments have to do with the basic details of the graph.

1. Thou shalt pay very close attention to thy axes, for therein lieth great opportunity to succeed or to fail

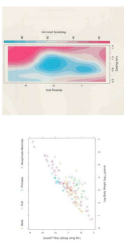
Focus on unit adjustments, tick intervals, eliminating unhelpful scale breaks, and manipulating axis lengths to correctly display data distribution.

Figure 1 shows a graph with a continuous scale, while Figure 2 shows a graph with an uneven scale making the relationship in the data appear more dramatic.



2. Thou shalt use color to categorize, not accessorize

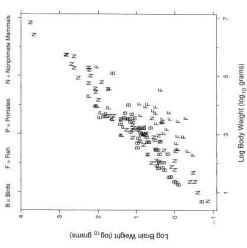
We should try to avoid using color in your graphs as just an accessory, but instead enhance the visual decoding of the information in the data. There are only two uses of color that transmit useful information: to encode categorical data to provide efficient assessment of each category as a whole, visually filtering out the other categories based on their different colors, and to encode the values of a quantitative numerical function of two factor variables by a color level plot, more frequently known as a contour plot.



3. Thou shalt choose symbols that can be easily distinguished from one another

Use symbols with distinct visual 'textures' to prevent confusion when data points overlap.

This figure shows a graph with textures that may be difficult to distinguish.



Minimizing 'chartjunk' and

maximizing 'data-ink'

The next three commandments focus on what you should maximize and minimize in your chart and how to display the data properly.

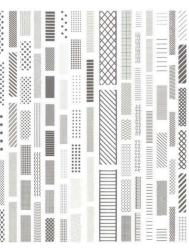
4. Thou shalt not employ 'chartjunk.'

You want to avoid:

1. Unintentional optical art

The moiré effect describes the phenomenon when the graphic design interacts with the physiological tremor of the eye to generate the distracting appearance of vibration and movement. This adds unnecessary noise to the decoding process and is particularly prominent in the "hill" for bar charts.

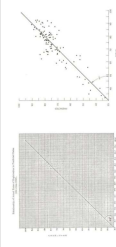
This figure gives examples of textures that exhibit the moiré effect.



2. The dreaded grid

This is when the purpose of grid lines, when used at all, should be to enhance the reading of the data. Dark grid lines enter the realm of chartjunk by competing with the data for the attention of the viewer. Thus, they should be muted or completely suppressed.

Figure 1 shows how grid lines can make the data almost illegible. Figure 2 shows a much clearer graph where the grid lines aren't a problem.



3. The self-promoting graphical duck!

Here the focus of the graphic is not to help interpret the information in the data, but instead every element of the graph is used to be more visually appealing instead of to tell the story of the data.

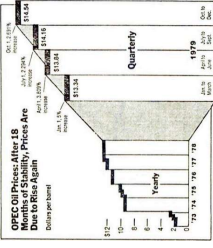
When a graphic is taken over by decorative forms or computer debris, when the data measures and structures become Design Elements, when the overall design purveys Graphical Style rather than quantitative information, then that graphic may be called a duck in honor of the duck-form store, "Big Duck."



5. Thou shalt show variation in thy data, not in thy design.

Data in the real world is going to have variation, and this is what you want to showcase with clarity. When changes in design are used, the reader may confuse the change in design with changes in data.

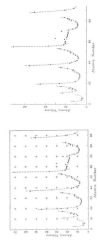
This Figure shows how an inconsistent design can lead to a misrepresentation of data.



6. Thou shalt maximize the data-ink ratio in thy graphs.

The data-ink ratio is quite simply the amount of "ink" used to depict the actual data divided by the total "ink" used to print the graphic. In other words, you want to waste less ink on gratuitous decoration and more on the actual representation of the data.

Figure 1 shows a lower data-ink ratio of 0.6, as there are unnecessary background dashes, while Figure 2 shows a much higher data-ink ratio of 0.9 which is much more effective.



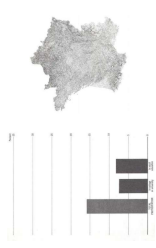
Designing for the Viewer

The final four commandments have to do with designing the graphic with the viewer in mind. If the viewer doesn't understand the information put forth in your graphic, it is ultimately a failure.

7. Thou shalt maximize thy data density and the size of thy data matrix (within reason).

The human eye has the ability to detect large amounts of information in small spaces, and we should take advantage of that ability by seeking to maximize the data density in our graphics. It is important however to keep this density within reason so that the information is still discernible.

Figure 1 is an extreme example of low data density, while Figure 2 shows the location and boundaries of 30,000 French communes. This is an example of too much data density.



8. Thou shalt draw the viewer's eye to the data, not to other design elements.

The fundamental principle of good statistical graphics is: above all else show the data. This means the sole focus should be on how the data is presented and not on the design elements. You should avoid adding an element if it doesn't contribute to the story the data is telling, or if it makes the data itself harder to understand.

9. Thou shalt do and redo thy graphs to determine which one telleth the story best.

Never fear to revise and edit your graphic design in a search for that which communicates most clearly to the largest audience. It would be good to remember that this process is a complex and multivariate one in which not only efficiency, but complexity, structure, density, and even beauty have a role to play in the generation of the final product.

10. Thou shalt not create "unfriendly," but "friendly" data graphics.

The creator of a graphic should try to keep in mind the viewer of his creation and not lost in his own world of graphical elements.

Here is a graphic that has some general tips on how to do this:

1. Use a clear, simple title that states the purpose of the graphic.	2. Use a clear, simple legend that identifies the data series.
3. Use a clear, simple axis label that states the unit of measurement.	4. Use a clear, simple axis scale that is easy to read.
5. Use a clear, simple axis tick mark that is easy to read.	6. Use a clear, simple axis grid that is easy to read.
7. Use a clear, simple axis break that is easy to read.	8. Use a clear, simple axis label that states the unit of measurement.
9. Use a clear, simple axis scale that is easy to read.	10. Use a clear, simple axis tick mark that is easy to read.

Conclusion

In conclusion, "The Elements of Graphing Data Well" successfully bridges the gap between sophisticated data visualization software and the core psychological principles of human perception. By synthesizing the timeless design philosophies of Cleveland and Tufte into an actionable framework, the article provides data creators with a practical toolkit to eliminate "chartjunk," maintain strict data integrity, and design with empathetic clarity for the audience. Ultimately, the paper accomplishes a vital goal for modern scientific and technical communication: it moves practitioners away from thoughtless, software-driven decoration and guides them toward creating elegant, honest, and highly efficient visual displays that reveal the true stories hidden within complex data.